Protocol

Systematic review and meta-analysis of mental health effects of pandemic in people with pre-existing mental illness

Background

The effect of Pandemics on the mental health of general population is widely published. Clinical services supporting people with mental illness are struggling to continue service provision in out-patient and in-patient settings. It is unclear as to how the pandemic affects people with pre-existing mental illness. Despite extensive Covid-19 related researches very little focus on how pandemics affect those with pre-existing mental illness.

Review questions

To conduct a systematic review and meta-analysis of existing literature to answer the following questions:
a. Do people with pre-existing mental illness experience an increase in mental health symptoms during a pandemic?
b. Do people with pre-existing mental illness experience more hospitalizations during a pandemic?

Methods

To iteratively develop a search strategy that will be run in the Medline, PsychInfo and Embase databases. Search strategy to capture known pandemics and all encompassing psychiatric illnesses.
The search is to be conducted from inception dates of the databases.

Inclusion criteria:

- Study reporting original data on people with pre-existing mental illness, assessing the effect of pandemic as its main objective. Study exploring psychiatric symptoms and/or hospitalizations.
- Providing quantitative data.

Exclusion criteria:

- Studies only describing mental health in people without mental illness
- Not assessing effects of known pandemic as recognized by WHO
- Not reporting on quantitative data
- Editorials, letters, commentaries, reviews not reporting original data

Retrieved articles to be exported into Endnote X9 Software where it will be screened independently by NA and JN following the stages of PRISMA flowdiagram. Articles that meet eligibility will be identified and grouped in a single folder for further evaluation. Where there is a disagreement on an article to include or exclude, a third reviewer KN will be involved.

Data will be extracted independently by NA and JN from all included studies. A proforma for extraction will be used including the items: study name, country of origin, type of pandemic, study design, description of participants and control group, type of mental illness, details of any rating scales, quantitative data in the form of means, sample size, standard deviations, percentages or prevalence rates or proportions.

RoBANS (risk of bias) will be independently completed by KN and VD. A consensus on score will be reached following discussion and when needed involvement of SL.

Strategy for data synthesis

The systematic review will include a narrative synthesis by examining all included studies in detail. The narrative will be described under the headings of the review questions. Study design, mental disorders, pandemic type and key findings will be presented to answer the review questions.

Meta-analysis will be performed on studies that report means, standard deviations and sample size in people with pre-existing mental illness and control group. The Comprehensive Meta-Analysis version 2 will be used to perform meta-analysis and to generate a standardized mean difference as the effect size. A random-effects model will be used. Heterogeneity will be measured using the I-squared and p-value and variance in effect size will be estimated by using tau-squared. Sensitivity analysis will be performed removing each study to see its impact and also on the type of pandemic, country of origin and mean age. As the purpose of the study is to examine if people with pre-existing mental illness experience increase in mental health symptoms we combined all reported mental health symptoms. This is to understand the magnitude of the impact on their overall mental health. Predominate symptoms from rating scales will be pooled to enable to categorize the effects of pandemic on particular symptom domains. Publication bias will be initially assessed using a Fail-safe N approach.

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